

lagCAMB

Beyond- Λ CDM Models for CAMB

User Manual

Dark Matter Interactions • Modified Gravity
Interacting Dark Energy • Ultralight Axions
Warm/Decaying Dark Matter

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Based on CAMB (Code for Anisotropies in the Microwave Background)

A. Lewis, A. Challinor & A. Lasenby (2000)

Contents

1	Introduction	2
1.1	Quick Start	2
1.2	Cosmological Baseline	2
1.3	Model Summary	2
2	Dark Matter Models	3
2.1	DM–Baryon Scattering	3
2.2	DM–Dark Radiation (ETHOS)	5
2.3	Decaying Dark Matter	7
2.4	DM–Neutrino Scattering	9
2.5	Fuzzy Dark Matter	11
2.6	Warm Dark Matter	13
2.7	DM–Photon Scattering	15
2.8	Multi-Channel Interacting DM	17
2.9	ETHOS Transfer (Murgia α – β – γ fit)	19
2.10	ETHOS Transfer (Physical surrogate)	21
3	Dark Energy & Modified Gravity Models	23
3.1	Interacting Dark Energy	23
3.2	Horndeski Scalar-Tensor Gravity	25
3.3	μ – Σ Modified Gravity	27
3.4	Fuzzy DM Field (Klein–Gordon Solver)	29
4	Advanced Usage	31
4.1	Combining DM and DE Models	31
4.2	Computing Derived Quantities	31
4.3	Recovering Λ CDM	31
5	Technical Notes	33
5.1	Fortran–Python Interface	33
5.2	CAMB Density Convention	33
5.3	Index Allocation	33
5.4	Numerical Approximations	33
6	References	33

1 Introduction

lagCAMB extends the standard CAMB Boltzmann solver with a modular framework for beyond- Λ CDM physics. All new models are implemented as plug-in classes inheriting from `TDarkMatterModel` (dark matter sector) or `TDarkEnergyModel` (dark energy / modified gravity), with minimal modifications to the core solver.

1.1 Quick Start

All models follow the same pattern:

```
import camb
from camb.dark_matter import DMBaryonScattering  # or any model

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)

# Activate a dark matter model
pars.DarkMatter = DMBaryonScattering()
pars.DarkMatter.set_params(sigma_dmb=1e-41, n_dmb=0, m_dm=1.0)

results = camb.get_results(pars)
cls = results.get_cmb_power_spectra(CMB_unit='muK')['total']
```

1.2 Cosmological Baseline

Unless otherwise noted, all comparison plots use the following baseline cosmology:

Parameter	Value	Description
H_0	67.5 km/s/Mpc	Hubble constant
$\Omega_b h^2$	0.022	Baryon density
$\Omega_c h^2$	0.122	CDM density
A_s	2.1×10^{-9}	Scalar amplitude
n_s	0.965	Spectral index
$\ell_{\max}$	2500	Multipole truncation

1.3 Model Summary

Model	Type	Primary Effect	Reference
DM–Baryon	DM	CDM–baryon drag	Boddy & Gluscevic 2018
DM–DR (ETHOS)	DM	DM–dark radiation coupling	Cyr-Racine+ 2016
Decaying DM	DM	CDM decay to daughter + DR	Blackadder+ 2014
DM–Neutrino	DM	CDM–neutrino scattering	He+ 2023
Fuzzy DM	DM	Quantum Jeans suppression	Hu+ 2000
Warm DM	DM	Free-streaming suppression	Viel+ 2005
DM–Photon	DM	CDM–photon opacity	Boddy & Gluscevic 2018
Multi-IDM	DM	Combined channels	Becker+ 2021
ETHOS Transfer (Murgia)	DM	Small-scale $T(k)$ fit	Murgia+ 2017
ETHOS Transfer (Physical)	DM	ETHOS params $\rightarrow T(k)$	Cyr-Racine+ 2016
Interacting DE	DE	DM–DE energy exchange	Costa+ 2017
Horndeski	DE/MG	Scalar-tensor gravity (QSA)	Bellini & Sawicki 2014
μ - Σ MG	MG	Phenomenological MG	Zhao+ 2009

2 Dark Matter Models

2.1 DM–Baryon Scattering

Physics

Dark matter scatters off baryons with a velocity-dependent momentum-transfer cross section:

$$\sigma_{\text{MT}} = \sigma_0 \left(\frac{v}{c} \right)^n \quad (1)$$

where $n \in \{-4, -2, 0, 2, 4\}$. This adds a drag force between the CDM and baryon fluids, suppressing small-scale structure and modifying the CMB damping tail. The back-reaction on baryons is included via momentum conservation.

Parameters

Name	Type	Default	Description
sigma_dmb	float	0.0	Cross section σ_0 [cm ²]
n_dmb	int	0	Velocity index $(-4, -2, 0, 2, 4)$
m_dm	float	100.0	DM particle mass [GeV]

```
from camb.dark_matter import DMBaryonScattering

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = DMBaryonScattering()
pars.DarkMatter.set_params(sigma_dmb=1e-41, n_dmb=0, m_dm=1.0)
# n_dmb=0: constant cross section
# Lighter DM (m_dm=1 GeV) gives stronger effect

results = camb.get_results(pars)
```

DM-Baryon Scattering

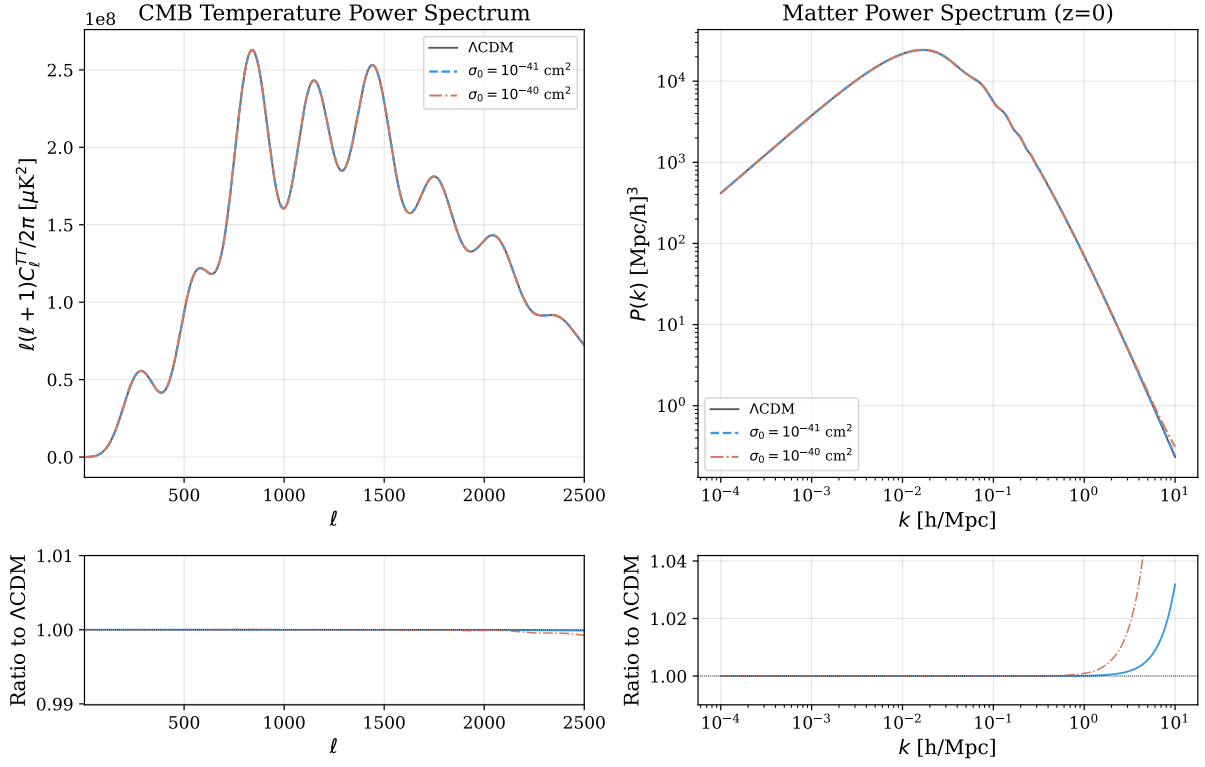


Figure 1: DM-Baryon scattering: CMB TT power spectrum and matter power spectrum compared to Λ CDM. Parameters: $n = 0$ (constant cross section), $m_{\text{DM}} = 1 \text{ GeV}$.

Note

Drag rates are internally capped at $10^4 \times \dot{a}/a$ to prevent ODE stiffness at very early times. For $n_{\text{dmb}} = -4$ (Coulomb-like), the effect is strongest at low velocities (late times).

2.2 DM–Dark Radiation (ETHOS)

Physics

In the ETHOS (Effective Theory of Structure Formation) framework, dark matter couples to a dark radiation sector via scattering. DM evolves as a fluid ($\delta_{\text{DM}}, v_{\text{DM}}$) while DR is described by a Boltzmann hierarchy ($F_0, F_1, \dots, F_{\ell_{\text{max}}}$). The opacity is parameterized as:

$$\dot{\kappa} = \sum_{n=2}^6 a_n \left(\frac{T_{\text{DR}}}{T_0} \right)^n \quad (2)$$

The interaction produces dark acoustic oscillations (DAO) analogous to baryon acoustic oscillations. Three regimes are automatically handled: tight coupling (TCA, $\Gamma \gg k$), full Boltzmann hierarchy, and ultra-relativistic fluid approximation (UFA, $k\tau \gg 1$).

Parameters

Name	Type	Default	Description
omdmrh2	float	0.0	Interacting DM density $\Omega_{\text{dmr}} h^2$
N_dark	float	0.0	DR effective number ΔN_{eff}
a_dark	list	None	ETHOS opacity coefficients $[a_2, \dots, a_6]$
alpha_l_dark	float	1.0	Angular damping for $\ell \geq 2$
lmax_dr	int	15	DR hierarchy truncation
cs2_dm	float	0.0	DM sound speed squared

```
from camb.dark_matter import DMDR_ETHOS

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = DMDR_ETHOS()
pars.DarkMatter.set_params(
    omdmrh2=0.005,          # 5% of CDM is interacting
    N_dark=1.0,             # 1 DR species
    a_dark=[1e5],           # strong coupling
    lmax_dr=15              # Boltzmann hierarchy depth
)
results = camb.get_results(pars)
```

DM-Dark Radiation (ETHOS)

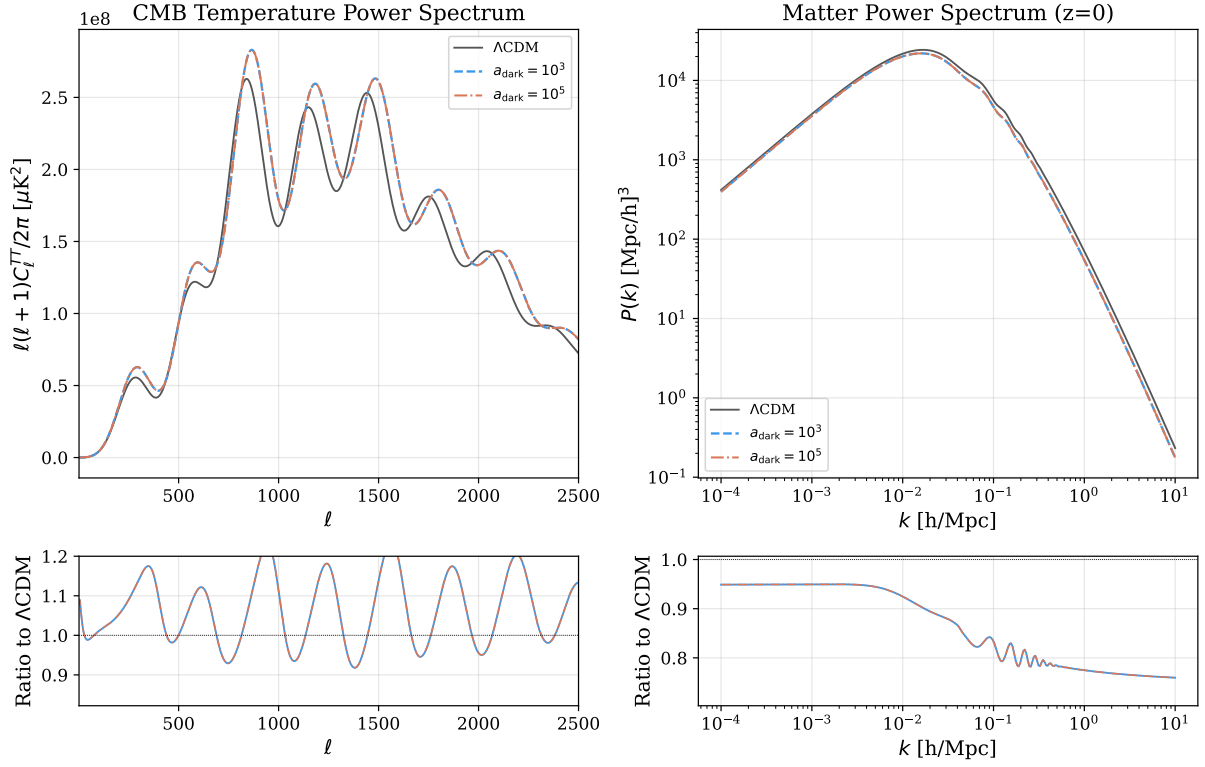


Figure 2: DM-DR (ETHOS): Dark acoustic oscillations appear in both CMB and $P(k)$ at scales entering the horizon before DM-DR decoupling. Stronger coupling ($a_{\text{dark}} = 10^5$) produces deeper oscillations.

Note

The implementation uses a tight-coupling approximation (TCA) when the DM-DR interaction rate exceeds $50k$, eliminating stiff coupling terms from the ODE system. This allows stable integration even for very large a_{dark} values ($\sim 10^7$).

2.3 Decaying Dark Matter

Physics

A fraction f_{dcdm} of CDM decays into a massive daughter particle and dark radiation:

$$\text{DM}_{\text{parent}} \rightarrow \text{DM}_{\text{daughter}} (\text{mass fraction } \epsilon) + \text{DR} \quad (3)$$

At decay (radial 2-body kinematics) the daughter gets a kick $v_{\text{kick}} = (1 - \epsilon^2)/(2\epsilon)$ and DR carries the remaining $(1 - \epsilon^2)/2$ of the rest energy. Daughter momentum redshifts as $p_d \propto 1/a$, so the daughter's effective equation of state evolves as

$$w_{\text{eff}}(a) = \min\left(\frac{1}{3}, w_{\text{kick}}(a_*/a)^2\right), \quad w_{\text{kick}} = \frac{1}{3}v_{\text{kick}}^2, \quad (4)$$

with $a_* = (3H_0/2\Gamma)^{2/3}$ the typical decay scale factor. The parent decays as $\rho_p(a) \propto a^{-3}e^{-\Gamma t(a)}$. The daughter is evolved as a Hu 1998 GDM warm fluid: δ_d and v_d obey

$$\dot{\delta}_d = -(1 + w_{\text{eff}})(kv_d + kz) - 3\mathcal{H}(c_s^2 - w_{\text{eff}})\delta_d - \Gamma_a(\rho_p/\rho_d)(\delta_d - \delta_c), \quad (5)$$

$$\dot{v}_d = -\mathcal{H}(1 - 3c_s^2)v_d + \frac{c_s^2}{1 + w_{\text{eff}}}k\delta_d - \Gamma_a(\rho_p/\rho_d)v_d, \quad (6)$$

with sound speed $c_s^2 \simeq w_{\text{eff}}$. The DR is propagated through a truncated F_0, F_1 Boltzmann hierarchy with the daughter source. Drag rates are capped at $50\mathcal{H}$ for numerical stability.

Parameters

Name	Type	Default	Description
Gamma_dcdm	float	0.0	Decay rate [km/s/Mpc]
epsilon_dcdm	float	1.0	$m_{\text{daughter}}/m_{\text{parent}}$
f_dcdm	float	1.0	Fraction of CDM that decays

```
from camb.dark_matter import DecayingDM

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = DecayingDM()
pars.DarkMatter.set_params(
    Gamma_dcdm=100,          # decay rate ~ H_0
    epsilon_dcdm=0.9,        # 10% mass lost to DR per decay
    f_dcdm=0.1               # 10% of CDM decays
)
results = camb.get_results(pars)
```


Decaying Dark Matter

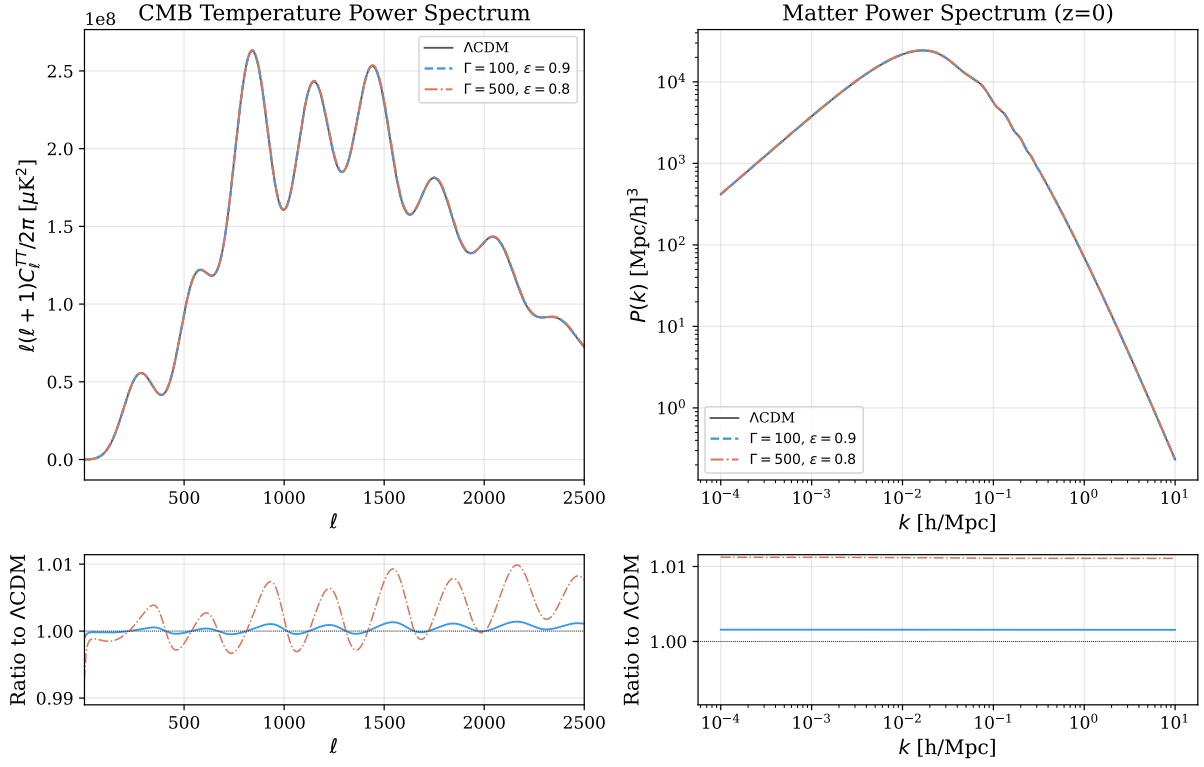


Figure 3: Decaying DM: Decay transfers CDM energy to relativistic DR, reducing matter density at late times. The daughter particle's velocity kick also contributes to small-scale suppression.

Note

The decay rate Γ is in the same units as H_0 [km/s/Mpc]. $\Gamma \sim H_0 \approx 67.5$ means the decay lifetime is comparable to the age of the universe. $\epsilon = 1$ corresponds to no mass loss (no decay effect).

2.4 DM–Neutrino Scattering

Physics

Dark matter scatters off neutrinos with an energy-dependent cross section:

$$\sigma = \sigma_0 \left(\frac{E_\nu}{1 \text{ MeV}} \right)^n \quad (7)$$

This adds drag to the CDM velocity equation and collision damping to the neutrino Boltzmann hierarchy, coupling CDM to the neutrino free-streaming scale.

Parameters

Name	Type	Default	Description
<code>sigma_dmnu</code>	float	0.0	Cross section at $E_0 = 1 \text{ MeV}$ [cm^2]
<code>n_dmnu</code>	int	0	Energy index (0 or 2)
<code>m_dm</code>	float	100.0	DM mass [GeV]

```
from camb.dark_matter import DMNeutrinoScattering

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = DMNeutrinoScattering()
pars.DarkMatter.set_params(
    sigma_dmnu=1e-33,      # cross section [cm^2]
    n_dmnu=0,              # constant cross section
    m_dm=0.1               # 100 MeV DM
)
results = camb.get_results(pars)
```

DM-Neutrino Scattering

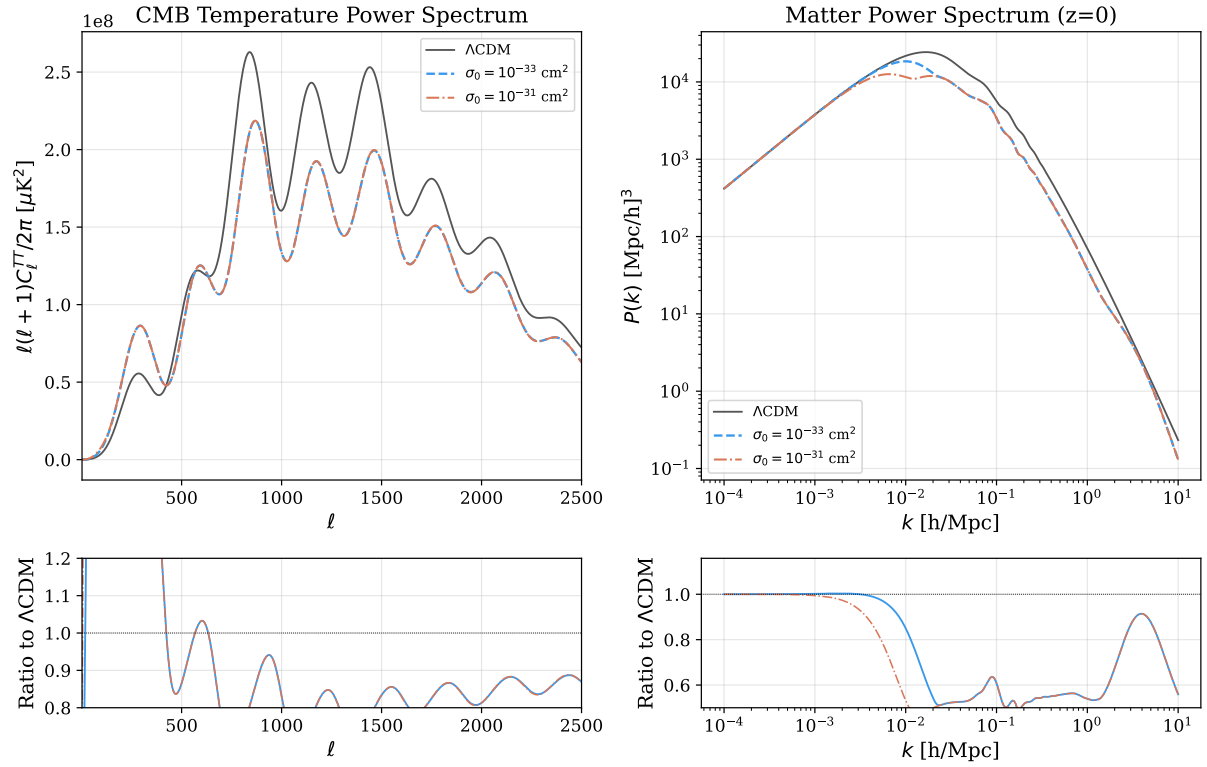


Figure 4: DM–Neutrino scattering: Coupling CDM to neutrinos delays matter domination on small scales and suppresses the matter power spectrum. Lighter DM gives a stronger per-particle scattering rate.

2.5 Fuzzy Dark Matter

Physics

Ultralight axion dark matter ($m \sim 10^{-22}$ eV) with de Broglie wavelength on astrophysical scales. The effective fluid has a scale-dependent sound speed:

$$c_s^2(k, a) = \frac{k^2}{4m^2 a^2} \quad (8)$$

producing a quantum Jeans scale $k_J^2 = 4m_{\text{conf}} a^2 \dot{a}/a$ below which density perturbations are suppressed by quantum pressure.

Parameters

Name	Type	Default	Description
m_axion	float	10^{-22}	Axion mass [eV]
omega_axion_h2	float	0.0	Axion density $\Omega_a h^2$ (0 uses f_axion)
f_axion	float	1.0	Fraction of CDM that is ULDM

```
from camb.dark_matter import FuzzyDM

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = FuzzyDM()
pars.DarkMatter.set_params(
    m_axion=1e-24,          # 10^{-24} eV
    f_axion=0.3             # 30% of CDM is ULDM
)
results = camb.get_results(pars)
```

Fuzzy Dark Matter

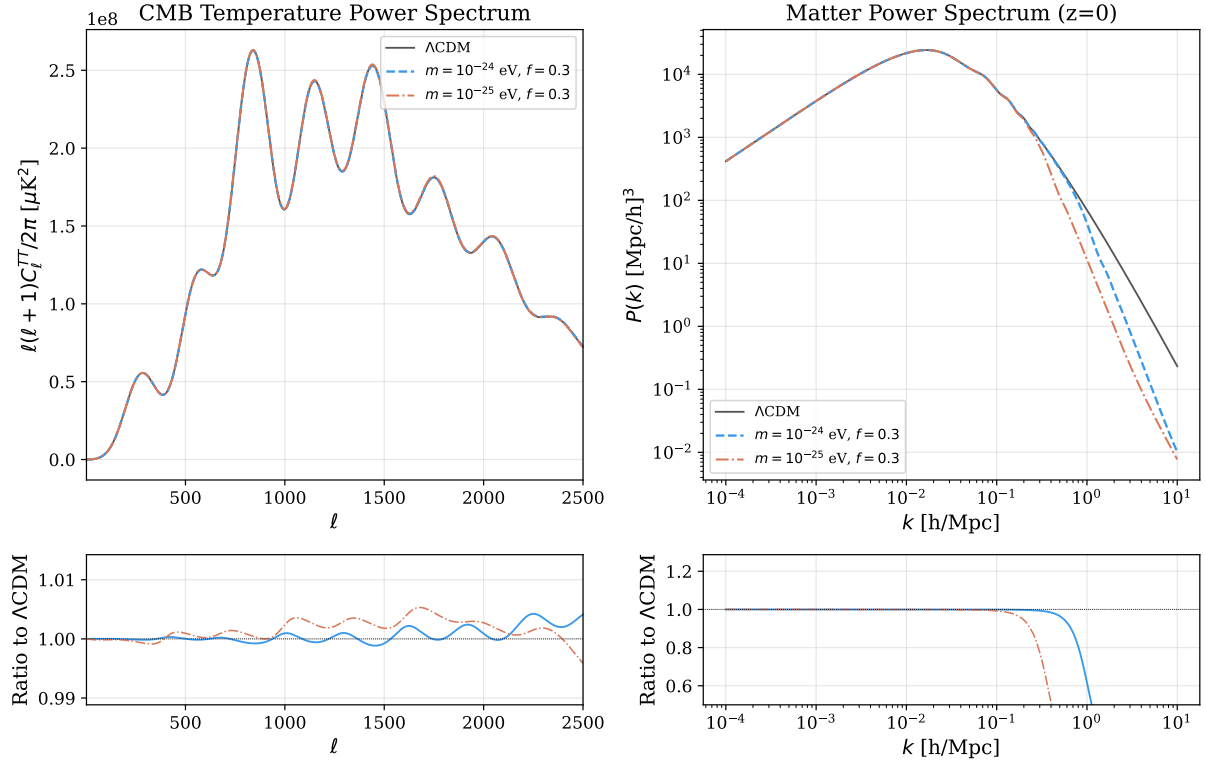


Figure 5: Fuzzy DM: Quantum pressure suppresses $P(k)$ below the Jeans scale, with smaller masses producing suppression at larger scales. The CMB is affected through the integrated Sachs–Wolfe effect.

Note

A full Klein–Gordon field implementation (**FuzzyDMField**) is also available in the DarkEnergy slot, solving the exact background evolution including the frozen-field era and Passaglia–Hu improved EFA for perturbations. See Section 3.4.

2.6 Warm Dark Matter

Physics

Thermal relic WDM with free-streaming that suppresses small-scale structure. In transfer-function mode (default):

$$T(k) = \left[1 + (\alpha k)^{2\nu}\right]^{-5/\nu} \quad (9)$$

where α depends on m_{WDM} , Ω_{WDM} , and h . In Boltzmann mode, WDM is treated as an extra massive neutrino species using CAMB's full neutrino hierarchy.

Parameters

Name	Type	Default	Description
m_wdm	float	3.0	WDM mass [keV]
T_wdm_ratio	float	1.0	T_{WDM}/T_ν ratio
Omega_wdm_h2	float	0.0	WDM density (0 = all CDM)
boltzmann_mode	bool	False	Full Boltzmann vs transfer function

```
from camb.dark_matter import WarmDM

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = WarmDM()
pars.DarkMatter.set_params(m_wdm=1.0) # 1 keV WDM
results = camb.get_results(pars)
```

Warm Dark Matter

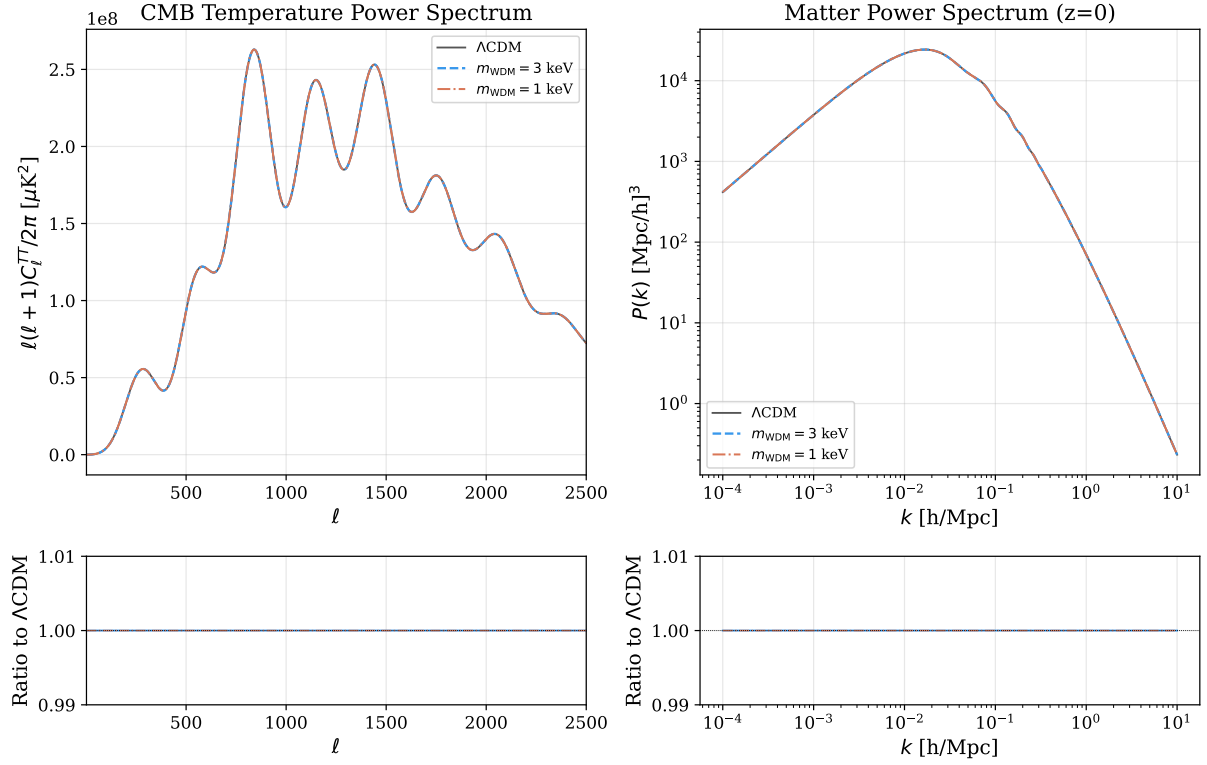


Figure 6: Warm DM: Free-streaming suppresses $P(k)$ at scales below the free-streaming horizon. Lighter particles ($m = 1$ keV) produce stronger suppression at larger scales.

Note

In transfer-function mode (default), the CMB power spectrum is unchanged from Λ CDM. Only the matter power spectrum is modified via the transfer function. Set `boltzmann_mode=True` to include CMB effects via the full Boltzmann hierarchy.

2.7 DM–Photon Scattering

Physics

DM scatters off photons with a temperature-dependent cross section:

$$\sigma = \sigma_{\text{Th}} u_{\text{idm-g}} \frac{m_{\text{DM}}}{100 \text{ GeV}} \left(\frac{T}{T_0} \right)^{n_{\text{idm-g}}} \quad (10)$$

This adds opacity to the photon hierarchy (extra scattering beyond Thomson) and drag to the CDM velocity.

Parameters

Name	Type	Default	Description
u_idm_g	float	0.0	Dimensionless coupling
n_idm_g	int	0	Temperature power index
m_dm	float	100.0	DM mass [GeV]

```
from camb.dark_matter import DMPHotonScattering

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = DMPHotonScattering()
pars.DarkMatter.set_params(u_idm_g=1e-4, n_idm_g=0, m_dm=100.)
results = camb.get_results(pars)
```


DM-Photon Scattering

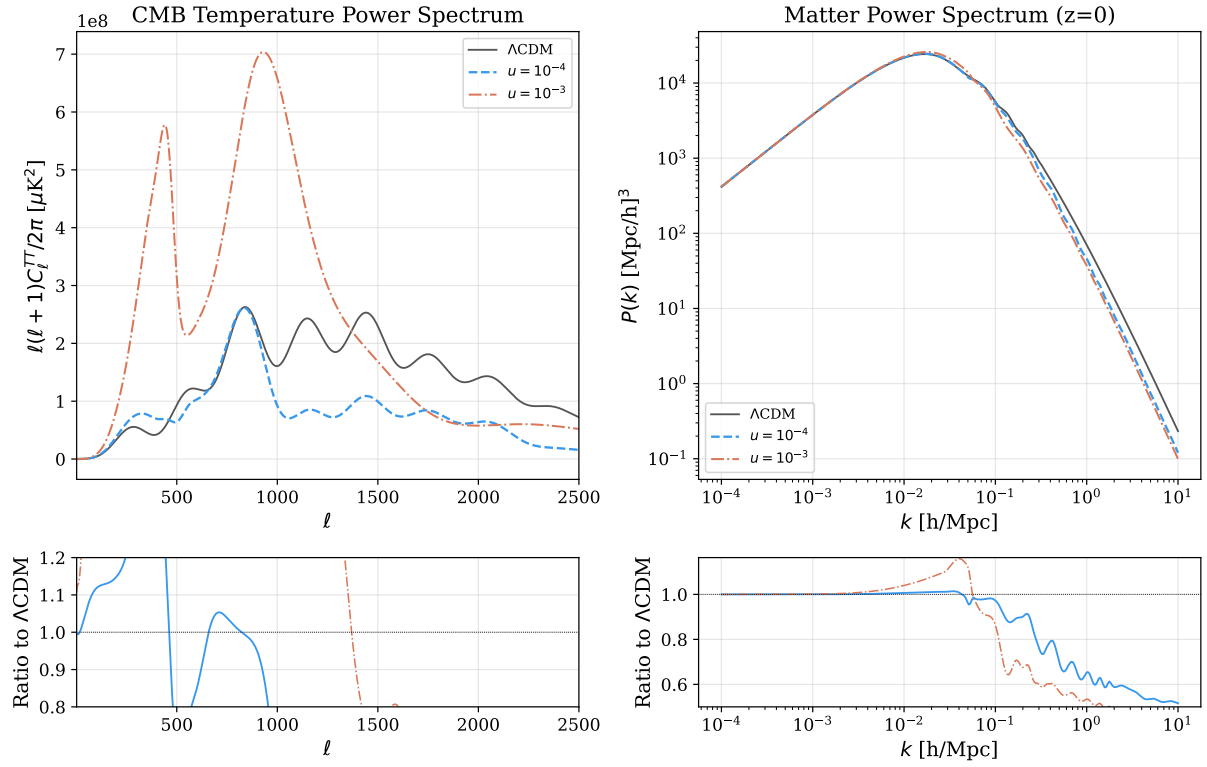


Figure 7: DM-Photon scattering: Extra photon opacity beyond Thomson scattering modifies Silk damping and suppresses small-scale power. Stronger coupling ($u = 10^{-3}$) shows pronounced CMB damping tail changes.

2.8 Multi-Channel Interacting DM

Physics

A single DM fluid with up to three simultaneous interaction channels:

- **Dark Radiation:** ETHOS-style DM–DR Boltzmann hierarchy
- **Baryon:** momentum-transfer scattering
- **Photon:** Thomson-like opacity

Channels activate independently based on parameter values, allowing combined studies of multiple DM interactions.

Parameters

All parameters from DMDR_ETHOS, DMBaryon, and DMPhoton are available:

Channel	Parameters
DR	omdmrh2, N_dark, a_dark, alpha_l_dark, lmax_dr
Baryon	sigma_dmb, n_dmb
Photon	u_idm_g, n_idm_g
Common	m_dm, cs2_dm

```
from camb.dark_matter import MultiInteractingDM

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = MultiInteractingDM()
pars.DarkMatter.set_params(
    # DR channel
    omdmrh2=0.005, N_dark=1.0, a_dark=[1e4],
    # Baryon channel
    sigma_dmb=1e-42, n_dmb=0,
    # Common
    m_dm=10.0
)
results = camb.get_results(pars)
```

Multi-Channel Interacting DM

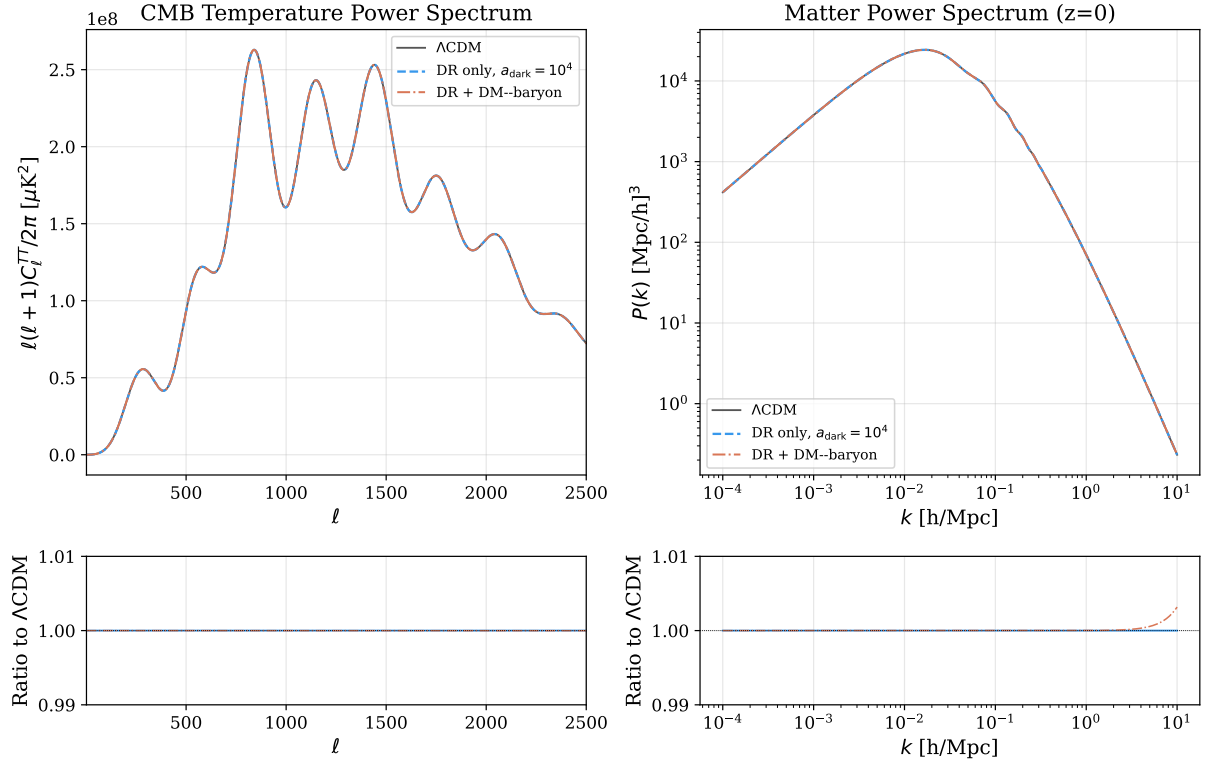


Figure 8: Multi-Channel IDM: combined DR (ETHOS) and DM-baryon channels. Adding a baryon-scattering channel on top of DR coupling produces additional small-scale suppression beyond the DAO bumps from the DR sector alone.

Note

Each channel contributes independently to the DM drag force. The number of perturbation equations varies dynamically: 1 (velocity only) to $1 + 1 + (\ell_{\text{max,DR}} + 1)$ when DR is active.

2.9 ETHOS Transfer (Murgia α - β - γ fit)

Physics

A fast analytic surrogate for ETHOS small-scale suppression of the matter power spectrum. The transfer function follows the Murgia et al. (2017) three-parameter fit:

$$T(k) = \left[1 + (\alpha k)^\beta\right]^\gamma \quad (11)$$

applied directly to $P(k) \rightarrow T(k)^2 P(k)$ and $\sigma_R \rightarrow T(k)\sigma_R$. The defaults $(\beta, \gamma) = (2.24, -4.46)$ match Murgia+ 2017 best-fit values; α [Mpc/h] sets the suppression scale. This model does NOT modify the CMB (no DR sector, no perturbations).

Parameters

Name	Type	Default	Description
alpha_mpch	float	0.0	Suppression scale α [Mpc/h] ($0 = \Lambda$ CDM)
beta_mur	float	2.24	Murgia exponent β
gamma_mur	float	-4.46	Murgia exponent γ

```
from camb.dark_matter import ETHOSTransferMurgia

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = ETHOSTransferMurgia()
pars.DarkMatter.set_params(alpha_mpch=0.05, beta_mur=2.24, gamma_mur=-4.46)
results = camb.get_results(pars)
```

ETHOS Transfer (Murgia)

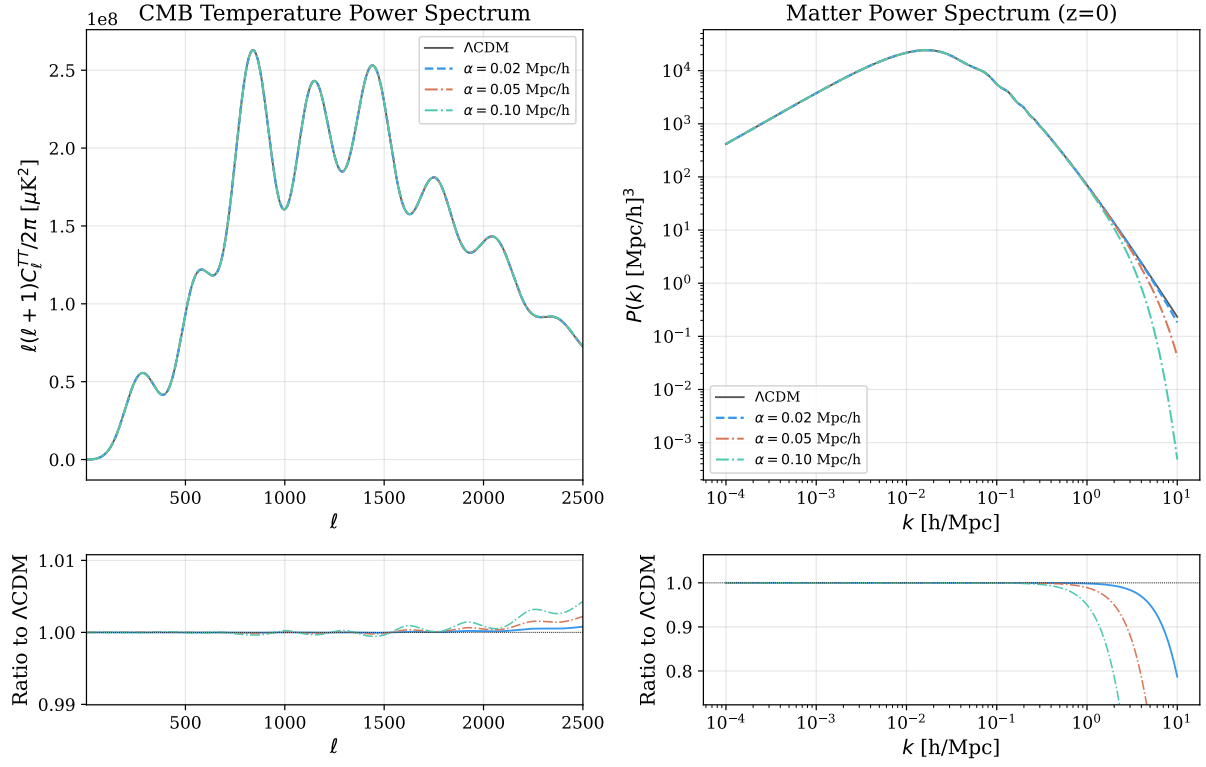


Figure 9: ETHOS Transfer (Murgia): The α - β - γ fitting form imposes a sharp small-scale cutoff in $P(k)$. Larger α moves the cutoff to larger scales. CMB is unchanged because the model only post-processes the matter transfer function.

Note

This is a fast surrogate for parameter scans / Lyman- α -style constraints; for the full self-consistent ETHOS solution (DAO + CMB effects) use `DMDR_ETHOS` (Section 2.2). For physical ETHOS parameters ($a_{\text{dark}}, \xi_{\text{DR}}, \dots$) mapped to α , see `ETHOSTransferPhysical` below.

2.10 ETHOS Transfer (Physical surrogate)

Physics

Same Murgia $T(k)$ form as above, but with α derived from physical ETHOS parameters via a power-law fit:

$$\alpha \text{ [Mpc/h]} = A_0 \left(\frac{a_{\text{dark,n}}}{a_{\text{ref}}} \right)^{p_a} \left(\frac{\xi_{\text{DR}}}{\xi_{\text{ref}}} \right)^{p_\xi} \left(\frac{\omega_{\text{dmdr}}}{0.12} \right)^{p_\omega} \quad (12)$$

with default $(A_0, p_a, p_\xi, p_\omega) = (0.5, 0.25, 1.0, -0.1)$ giving an order-of-magnitude correct mapping for $n_{\text{dark}} = 2$ (dark Thomson-like opacity). The fit constants are user-exposed so each science application can recalibrate against the full DMDR_ETHOS solution for its parameter ranges.

Parameters

Name	Type	Default	Description
a_dark_n	float	0.0	ETHOS opacity coefficient [Mpc ⁻¹]
xi_dr	float	0.5	T_{DR}/T_γ at recombination
omdmdrh2	float	0.12	Interacting DM density $\Omega_{\text{dmdr}} h^2$
n_dark	int	2	Opacity index (advisory)
A0_fit, a_ref, xi_ref	float	0.5, 1e-4, 0.5	Fit reference point
p_a, p_xi, p_om	float	0.25, 1.0, -0.1	Fit exponents
beta_mur, gamma_mur	float	2.24, -4.46	Murgia exponents

```
from camb.dark_matter import ETHOSTransferPhysical

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkMatter = ETHOSTransferPhysical()
pars.DarkMatter.set_params(a_dark_n=4e-4, xi_dr=0.5, omdmdrh2=0.12)
results = camb.get_results(pars)
```

ETHOS Transfer (Physical)

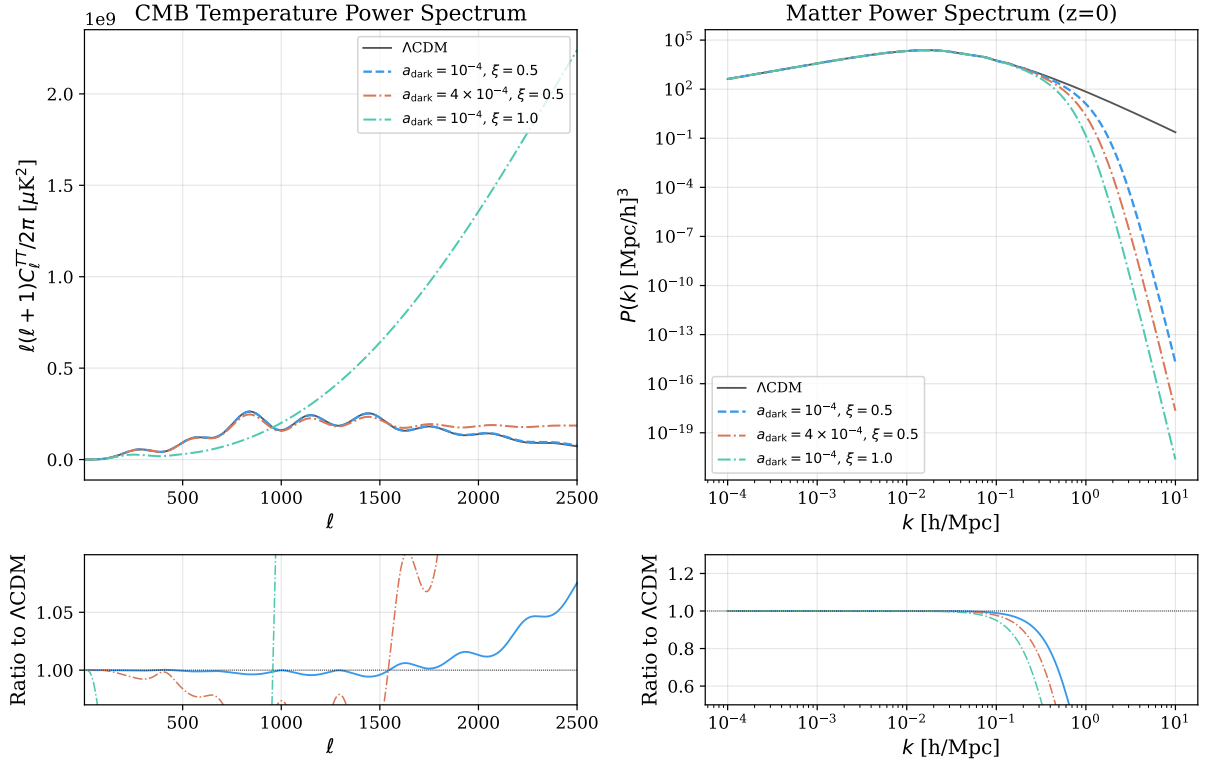


Figure 10: ETHOS Transfer (Physical): Increasing the ETHOS opacity $a_{\text{dark},n}$ or the dark-radiation temperature ratio ξ_{DR} pushes the suppression scale to larger scales. The default power-law mapping is intentionally a rough fit; recalibrate the $A_0, p_a, p_\xi, p_\omega$ exponents against `DMDR_ETHOS` for your parameter range.

Note

This is a parameter-scan surrogate. It does NOT reproduce dark acoustic oscillations or CMB effects — use `DMDR_ETHOS` (Section 2.2) for the full self-consistent calculation.

3 Dark Energy & Modified Gravity Models

3.1 Interacting Dark Energy

Physics

Dark matter and dark energy exchange energy-momentum Q^μ . The background conservation equations become:

$$\dot{\rho}_c + 3H\rho_c = Q, \quad \dot{\rho}_{\text{de}} + 3(1+w)H\rho_{\text{de}} = -Q \quad (13)$$

Three interaction kernels:

1. $Q = \xi H \rho_{\text{de}}$
2. $Q = \xi H \rho_c$
3. $Q = \xi H (\rho_c + \rho_{\text{de}})$

Perturbation equations are derived consistently from the energy-momentum tensor.

Parameters

Name	Type	Default	Description
w	float	-1.0	$w(0)$ equation of state
wa	float	0	$-dw/da(0)$
xi_ide	float	0.0	Coupling strength
interaction_type	int	1	Kernel type (1, 2, or 3)
cs2_ide	float	1.0	DE sound speed squared

```
from camb.dark_energy import InteractingDE

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkEnergy = InteractingDE()
pars.DarkEnergy.set_params(
    w=-0.9,
    xi_ide=0.05,
    interaction_type=1      # Q = xi * H * rho_de
)
results = camb.get_results(pars)
```


Interacting Dark Energy

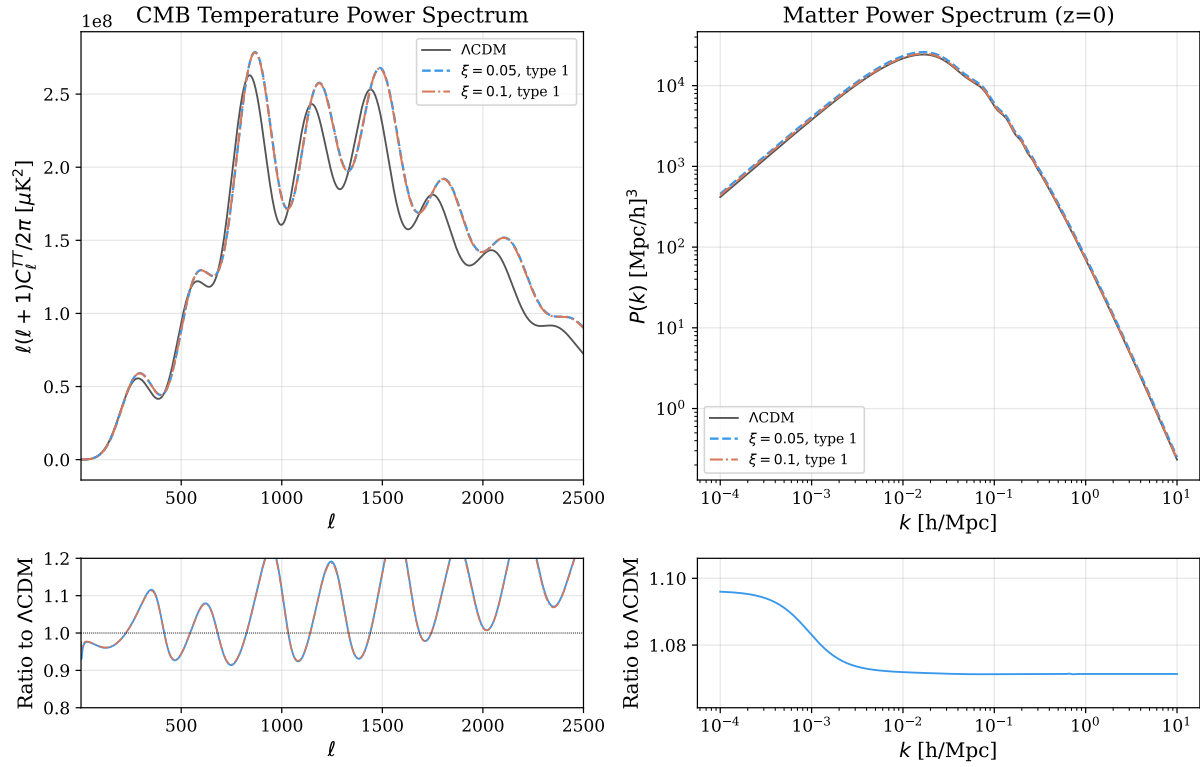


Figure 11: Interacting DE (type 1): Energy transfer from DE to DM modifies the expansion history and perturbation growth. The CMB is affected through the late ISW effect and modified distance to last scattering.

Note

Important: Always use `set_params()` as a single method call. Do not set individual attributes, because `pars.DarkEnergy` returns a new Python wrapper each time, so individual setters lose previous changes.

3.2 Horndeski Scalar-Tensor Gravity

Physics

The most general scalar-tensor theory with second-order equations of motion. Parameterized by four time-dependent α -functions (Bellini & Sawicki 2014):

- α_K : kineticity (kinetic energy of scalar)
- α_B : braiding (kinetic mixing with gravity)
- α_M : Planck mass run rate ($d \ln M_*^2 / d \ln a$)
- α_T : tensor speed excess ($c_T^2 = 1 + \alpha_T$)

Uses proportional parameterization: $\alpha_X(a) = \alpha_{X,0} \times \Omega_{\text{DE}}(a)$.

In the quasi-static approximation (QSA), these modify:

$$k^2 \Psi = -4\pi G \mu(a) a^2 \rho \Delta \quad (\text{Poisson}) \quad (14)$$

$$k^2 (\Phi + \Psi) = -8\pi G \Sigma(a) a^2 \rho \Delta \quad (\text{lensing}) \quad (15)$$

where μ and Σ are derived from the α -functions.

Parameters

Name	Type	Default	Description
w	float	-1.0	DE equation of state
wa	float	0	$-dw/da(0)$
alpha_K	float	0.0	Kineticity amplitude
alpha_B	float	0.0	Braiding amplitude
alpha_M	float	0.0	Planck mass run rate
alpha_T	float	0.0	Tensor speed excess
M_star_ini	float	1.0	Initial M_*^2/M_{Pl}^2

```
from camb.dark_energy import HorndeskiDE

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkEnergy = HorndeskiDE()
pars.DarkEnergy.set_params(alpha_M=0.1, M_star_ini=1.0)
results = camb.get_results(pars)
```

Horndeski Gravity

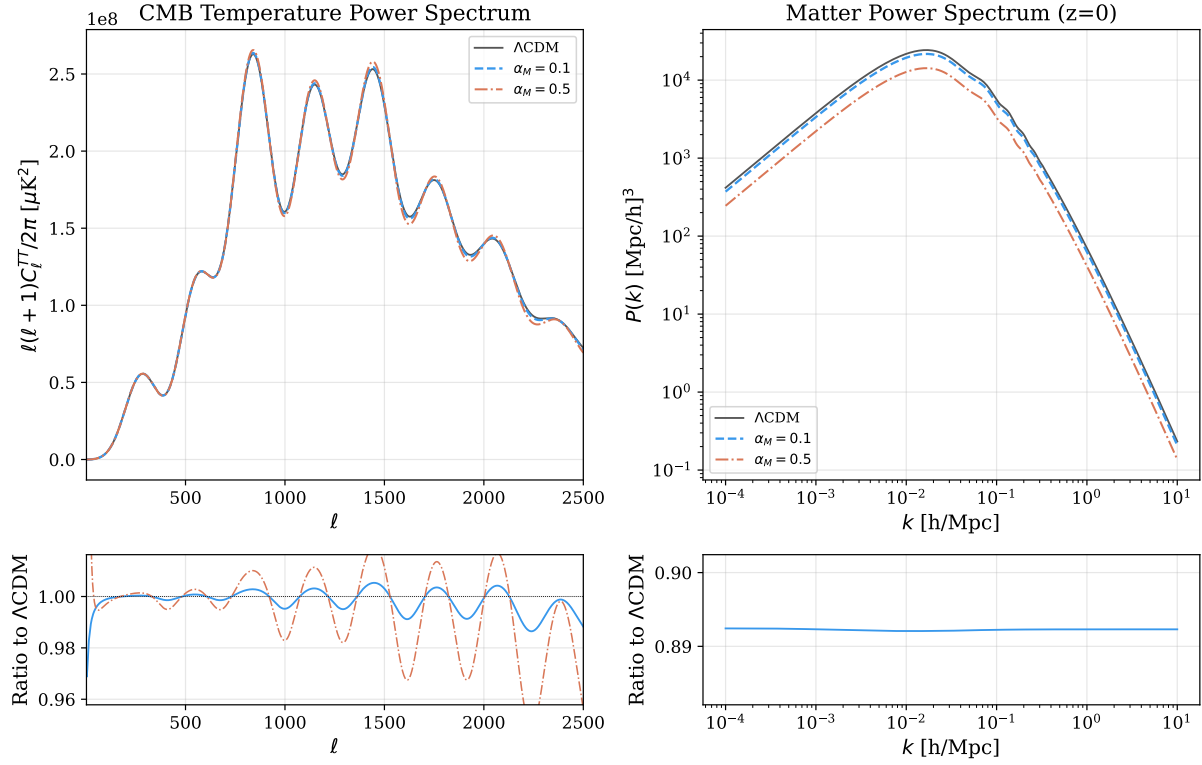


Figure 12: Horndeski gravity: α_M modifies the Planck mass evolution, affecting CMB lensing and the late ISW effect. The matter $P(k)$ is mostly unchanged in the current QSA implementation (see note below).

Note

Known limitation (QSA): In the current implementation, the Horndeski QSA only modifies the lensing (Weyl) potential and tensor propagation. Matter growth (σ_8) is *not* modified because in synchronous gauge, the μ -modification affects the first derivative of δ_c rather than the second, giving incorrect growth for time-dependent μ . A constant M_*^2 (`M_star_ini` $\neq 1$) works correctly through accumulated state change. Full scalar field perturbation equations are needed for proper growth modification.

3.3 μ - Σ Modified Gravity

Physics

Phenomenological parameterization of deviations from GR in the Poisson and lensing equations:

$$k^2 \Psi = -4\pi G \mu(a, k) a^2 \rho \Delta \quad (16)$$

$$k^2 (\Phi + \Psi) = -8\pi G \Sigma(a, k) a^2 \rho \Delta \quad (17)$$

with $\mu = 1 + \mu_0 \Omega_{\text{DE}}(a)$ and $\Sigma = 1 + \Sigma_0 \Omega_{\text{DE}}(a)$. Optional scale-dependent modification via $\lambda_\mu, \lambda_\Sigma$ [Mpc]. In GR: $\mu = \Sigma = 1$.

Parameters

Name	Type	Default	Description
mu_0	float	0.0	μ deviation amplitude
sigma_0	float	0.0	Σ deviation amplitude
lambda_mu	float	0.0	μ scale [Mpc] (0 = scale-indep.)
lambda_sigma	float	0.0	Σ scale [Mpc] (0 = scale-indep.)
c1	float	0.0	μ k -dependence coefficient
c2	float	0.0	Σ k -dependence coefficient

```
import camb

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

# Modified gravity via pars.MG (not DarkMatter/DarkEnergy)
pars.MG.is_active = True
pars.MG.mu_0 = 0.3
pars.MG.sigma_0 = 0.2
results = camb.get_results(pars)
```

Mu-Sigma Modified Gravity

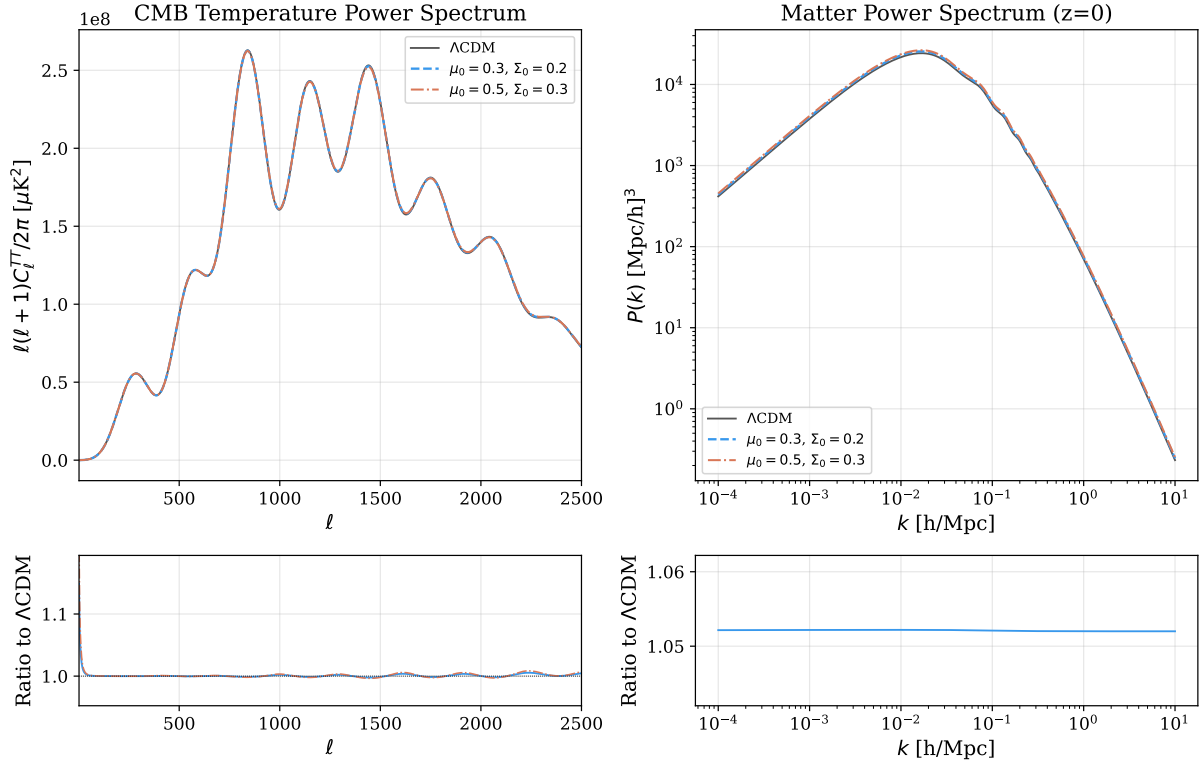


Figure 13: μ - Σ MG: Modified lensing and Poisson equations affect the CMB through the ISW effect and lensing. The matter $P(k)$ shows changes through the modified growth rate.

Note

μ - Σ MG is set via `pars.MG` (not `DarkMatter` or `DarkEnergy`). It is mutually exclusive with `HorndeskiDE`. This parameterization primarily modifies the Weyl potential (lensing) and has limited effect on the matter growth rate σ_8 in the current implementation.

3.4 Fuzzy DM Field (Klein–Gordon Solver)

Physics

Full Klein–Gordon background solver for ultralight axion dark matter, placed in the DarkEnergy slot. Solves:

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0, \quad V(\phi) = \frac{1}{2}m^2\phi^2 \quad (18)$$

At early times ($m \ll H$), the field is frozen at $w = -1$. After oscillation onset ($m \sim 3H$), the field oscillates rapidly and behaves as pressureless matter ($\langle w \rangle = 0$). The transition is handled by switching from exact KG to an effective fluid approximation (EFA) with the Passaglia–Hu sound speed:

$$c_s^2 = \frac{k^2}{4m^2a^2 + k^2} \quad (19)$$

Parameters

Name	Type	Default	Description
m_axion	float	10^{-22}	Axion mass [eV]
f_axion	float	0.0	Fraction of CDM+axion budget
omega_axion_h2	float	0.0	Axion density (0 uses f_axion)
N_match	int	100	KG→EFA criterion m/H
potential_type	int	1	1=quadratic, 2=cosine
n_potential	int	1	Power of cosine potential
f_decay	float	1.0	Decay constant [M_{Pl}]
use_improved_efa	bool	True	Passaglia–Hu improved EFA

```
from camb.dark_energy import FuzzyDMField

pars = camb.CAMBparams()
# Reduce omch2 to make room for axion component
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122*(1-0.05))
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0], kmax=10.0)

pars.DarkEnergy = FuzzyDMField()
pars.DarkEnergy.set_params(m_axion=1e-22, f_axion=0.05)
results = camb.get_results(pars)
```

Fuzzy DM Field

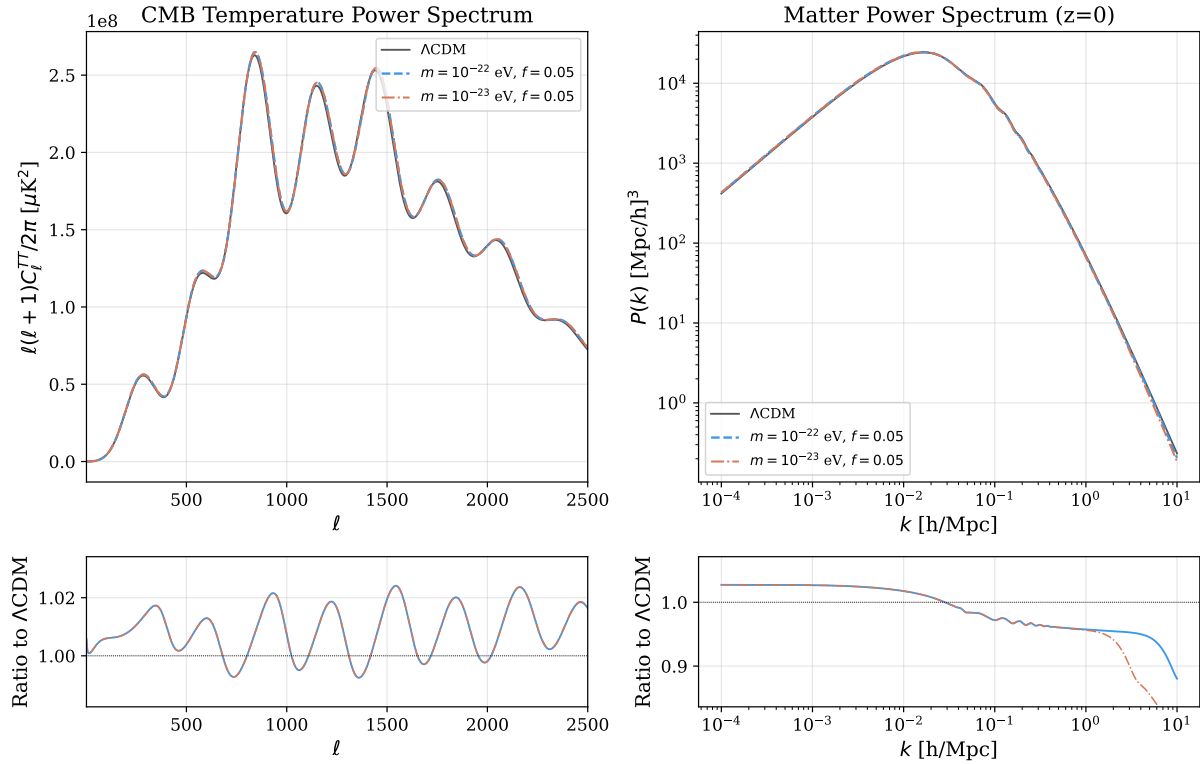


Figure 14: Fuzzy DM Field (Klein-Gordon): Lighter axion masses ($m = 10^{-23}$ eV) oscillate later, producing larger-scale Jeans suppression. The exact KG-to-EFA matching reproduces ultralight-axion CDM-like behavior at high k once oscillations begin.

Note

FuzzyDMField lives in the DarkEnergy slot (not DarkMatter), because it needs to control background evolution. You must manually reduce `omch2` by the axion fraction: `omch2 *= (1 - f_axion)`. The class cannot be pickled due to internal splines.

4 Advanced Usage

4.1 Combining DM and DE Models

DM and DE models occupy separate slots and can be combined freely:

```
from camb.dark_matter import DMBaryonScattering
from camb.dark_energy import InteractingDE

pars = camb.CAMBparams()
pars.set_cosmology(H0=67.5, ombh2=0.022, omch2=0.122)
pars.InitPower.set_params(As=2.1e-9, ns=0.965)
pars.set_for_lmax(2500, lens_potential_accuracy=1)

# DM sector: baryon scattering
pars.DarkMatter = DMBaryonScattering()
pars.DarkMatter.set_params(sigma_dmb=1e-42, n_dmb=0, m_dm=1.0)

# DE sector: interacting quintessence
pars.DarkEnergy = InteractingDE()
pars.DarkEnergy.set_params(w=-0.9, xi_ide=0.05)

results = camb.get_results(pars)
```

4.2 Computing Derived Quantities

```
# CMB power spectra
cls = results.get_cmb_power_spectra(CMB_unit='muK')
cl_tt = cls['total'][:, 0]
cl_ee = cls['total'][:, 1]
cl_te = cls['total'][:, 3]

# Matter power spectrum
pars.WantTransfer = True
pars.set_matter_power(redshifts=[0, 0.5, 1.0], kmax=10.0)
results = camb.get_results(pars)
kh, z, pk = results.get_matter_power_spectrum(
    minkh=1e-4, maxkh=10, npoints=300)

# sigma_8
sigma8 = results.get_sigma8_0()
print(f"sigma_8 = {sigma8:.4f}")

# Angular diameter distance
DA = results.angular_diameter_distance(1100)
print(f"D_A(z=1100) = {DA:.1f} Mpc")
```

4.3 Recovering Λ CDM

All models reduce to Λ CDM when their coupling parameters are set to zero:

```
# These are all equivalent to standard CAMB:
pars.DarkMatter = DMBaryonScattering()
pars.DarkMatter.set_params(sigma_dmb=0.0) # no scattering
```



```
pars.DarkMatter = WarmDM()  
pars.DarkMatter.set_params(m_wdm=100.0)      # very heavy = CDM  
pars.MG.is_active = False                    # GR
```

5 Technical Notes

5.1 Fortran–Python Interface

lagCAMB uses the same `ctypes` bridge as standard CAMB. Model parameters declared as `_fields_` in the Python class map to Fortran type fields at exact byte offsets. Models with Fortran `TCubicSpline` members (`InteractingDE`, `HorndeskiDE`) use `_methods_` to call Fortran setter subroutines instead.

5.2 CAMB Density Convention

Background densities use conformal-time units:

$$\text{grho*}_t = 8\pi G \rho a^2, \quad \text{grhoc}_t = \frac{\text{State}\% \text{grhoc}}{a} \quad (20)$$

5.3 Index Allocation

Perturbation variable indices in the ODE array `y(:)`:

Index	Variable	Description
<code>ix_clxc = 2</code>	δ_c	CDM density perturbation (fixed)
<code>EV%vc_ix</code>	v_c	CDM velocity (if DM model has it)
<code>EV%dm_ix</code>	δ_{DM} , etc.	DM extra equations
<code>EV%dr_ix</code>	$F_0, F_1, \dots$	DR Boltzmann hierarchy
<code>EV%w_ix</code>	$\delta_{\text{de}}, v_{\text{de}}$	DE perturbation equations

5.4 Numerical Approximations

- **TCA (Tight Coupling)**: When $\Gamma_{\text{DR}} + \Gamma_{\text{tilde}}/4 > 50k$, uses combined momentum equation to eliminate stiff collision terms.
- **UFA (Ultra-relativistic Fluid)**: When $k\tau \gg 1$, truncates DR hierarchy to 3 equations with arXiv:1104.2933 closure.
- **Drag caps**: All DM drag rates are capped at $10^4 \times \dot{a}/a$ to prevent ODE stiffness.

6 References

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